Community Solar Project Documentation Package

Document 1: Project Overview & Developer Information

SOUTHWEST RENEWABLE ENERGY PARTNERS, LLC
8900 Solar Park Boulevard, Suite 300
Tempe, AZ 85284
ROC License #: 298456 (Commercial Solar, Electrical Contractor)
Phone: (480) 555-0892
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Website: www.swrenewable.com

COMMUNITY SOLAR PROJECT PROPOSAL

Project Name: Desert Hills Community Solar Array
Date: September 20, 2025
Project ID: SWREP-CS-2025-14

Property Owner/Host Site:
City of Mesa - Parks & Recreation Department
20 East Main Street
Mesa, AZ 85201
Contact: David Wong, Facilities Director
Phone: (480) 555-0754
Email: david.wong@mesaaz.gov

Project Address:
Desert Hills Community Park
9845 East Brown Road
Mesa, AZ 85207
Parcel #: 301-22-045B
Zoning: Public/Institutional

Project Specifications

System Size: 750 kW DC / 600 kW AC
System Type: Ground-mounted, fixed-tilt array
Installation Type: Ballasted racking system on undeveloped park land

Annual Production Estimate: 1,425,000 kWh/year

Subscriber Capacity: Approximately 125 households (based on average 950 kWh/month usage)

Equipment Summary:

- **Solar Panels:** (2,000) Jinko Solar Tiger Neo N-type 375W bifacial modules
- **Inverters:** (6) SMA Sunny Central 100 CP XT central inverters (100 kW each)
- **Mounting:** Schletter FS Duo fixed-tilt ground mount system, 25° tilt, south-facing
- **Transformer:** (1) 750 kVA pad-mounted transformer, 480V to 12.47 kV
- **Monitoring:** SMA Data Manager with Sunny Portal
- **Perimeter Fencing:** 7-foot chain-link fence with barbed wire top, lockable gates
- **Access Road:** Gravel road from existing park road

System Coverage Area: Approximately 3.2 acres (footprint), 5.1 acres (total lease area)

Project Valuation: $1,425,000

Scope of Work

1. Site preparation and grading (minimal)
2. Installation of ballasted ground-mount racking system
3. Installation of 2,000 solar modules
4. Installation of inverters, combiner boxes, and electrical equipment
5. Installation of pad-mounted transformer
6. Trenching and conduit installation for electrical runs
7. Installation of perimeter fencing and signage
8. Utility interconnection at existing 12.47 kV distribution line
9. SCADA monitoring system installation
10. Vegetation management (native desert landscaping beneath/around array)

Project Timeline:

- Construction Duration: 16-20 weeks
- Estimated Start: January 2026 (post-permit approval)
- Commercial Operation: June 2026

Subscriber Program: Operated by Southwest Renewable Energy Partners in partnership with Salt River

Project (SRP). Subscribers receive credits on their SRP bills for their portion of solar energy production.

General Contractor:
Southwest Renewable Energy Partners, LLC
License #: ROC 298456
8900 Solar Park Boulevard, Suite 300
Tempe, AZ 85284
Project Manager: Jennifer Nakamura, PE
Phone: (480) 555-0892
Email: jnakamura@swrenewable.com

Electrical Contractor:
Desert Power Systems, Inc.
License #: E-98754 (Class A Electrical)
4521 Industrial Way
Phoenix, AZ 85034
Contact: Michael Torres
Phone: (602) 555-0445
Email: mtorres@desertpower.com

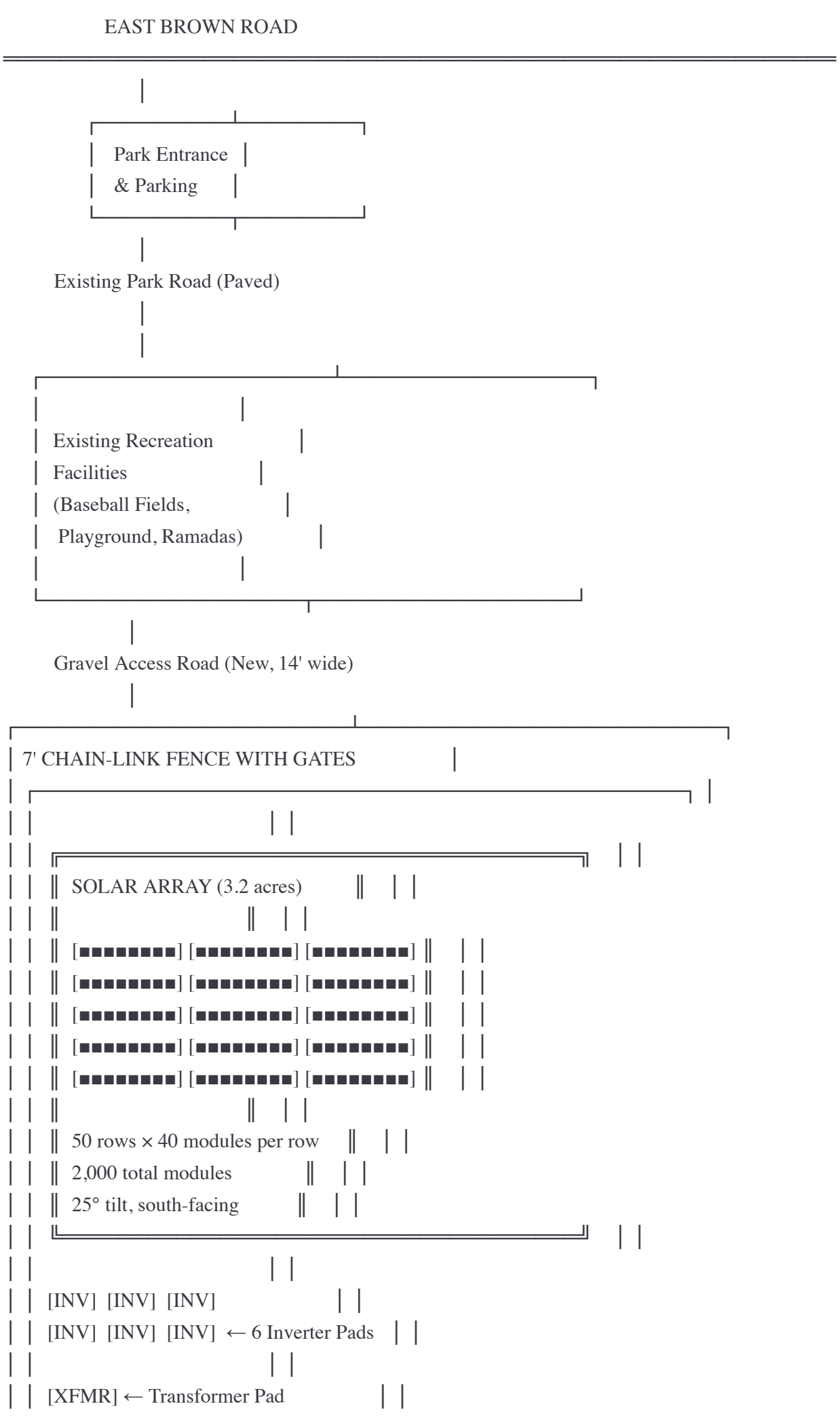
Civil Engineering:
Arizona Site Solutions, LLC
Professional Engineer: Sarah Mitchell, PE #47892
1200 Engineering Plaza
Scottsdale, AZ 85260
Phone: (480) 555-0231

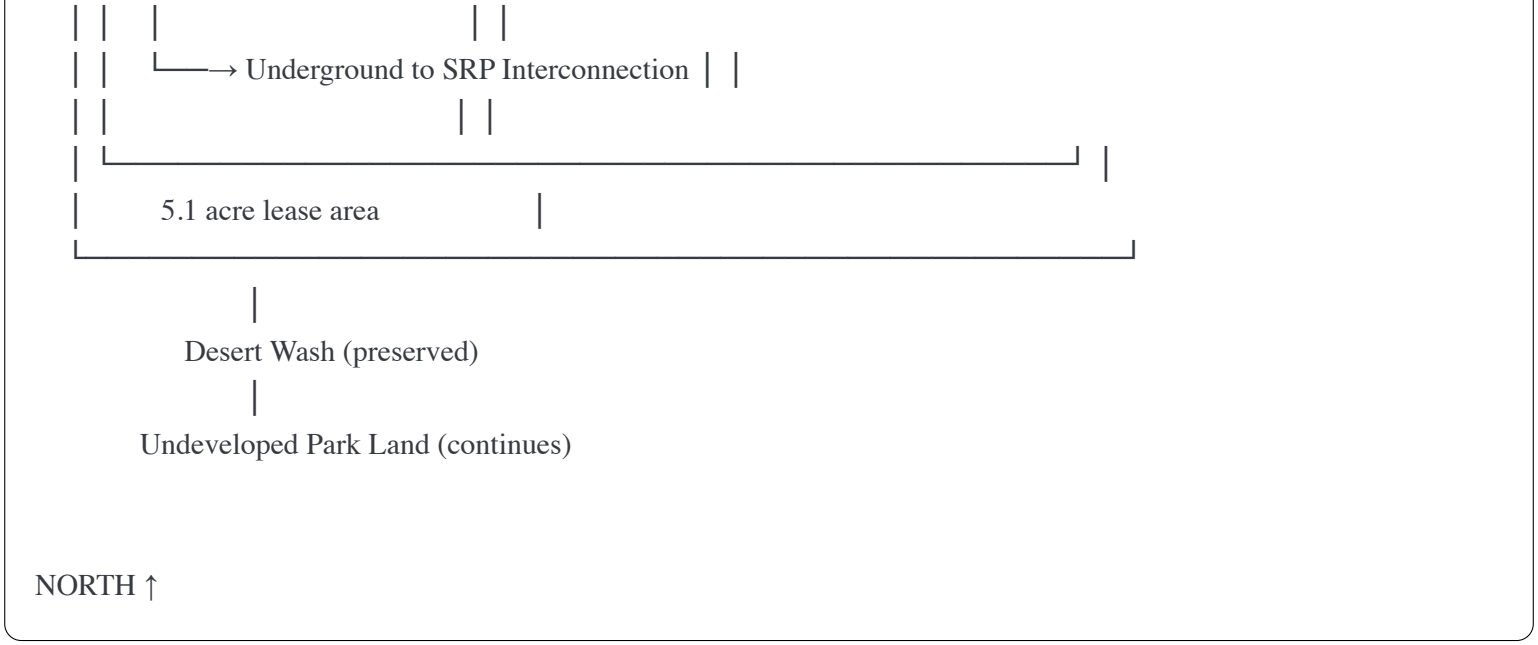
Document 2: Site Plan

SITE PLAN - DESERT HILLS COMMUNITY SOLAR ARRAY
9845 East Brown Road, Mesa, AZ 85207

Prepared by: Arizona Site Solutions, LLC
Date: September 15, 2025
Scale: 1" = 100'
Parcel Size: 47 acres (total park), 5.1 acres (solar lease area)

Site Layout


EAST BROWN ROAD
Park Entrance
& Parking
Existing Park Road (Paved)
Existing Recreation
Facilities
(Baseball Fields,
Playground, Ramadas)
Gravel Access Road (New, 14' wide)
7' CHAIN-LINK FENCE WITH GATES
SOLAR ARRAY (3.2 acres)
50 rows × 40 modules per row
2,000 total modules
25° tilt, south-facing
[INV] [INV] [INV]
[INV] [INV] [INV] ← 6 Inverter Pads
[XFMR] ← Transformer Pad



Key Features:

- **Array Configuration:** 50 rows, 40 modules per row, portrait orientation
- **Row Spacing:** 15 feet (center-to-center) to minimize shading
- **Setbacks:**
  - From fence line: 15 feet (all sides)
  - From property line: 25 feet (meets zoning requirement)
  - From desert wash: 50 feet (environmental buffer)
- **Equipment Locations:**
  - 6 inverter pads: Central location, 12' × 8' concrete pads
  - Transformer pad: 15' × 20' concrete pad with oil containment
  - Access gates: Main gate (20' wide) on north side, emergency gate (12' wide) on east side
- **Utility Interconnection:** Underground conduit runs 450 feet to existing SRP pole on East Brown Road
- **Drainage:** Natural sheet flow maintained, no significant grading required
- **Vegetation:** Native desert plantings (palo verde, brittlebush, etc.) for erosion control and visual screening

Coordinates (NAD83):

- Northwest corner: 33.4892°N, 111.6234°W
- Southeast corner: 33.4875°N, 111.6218°W

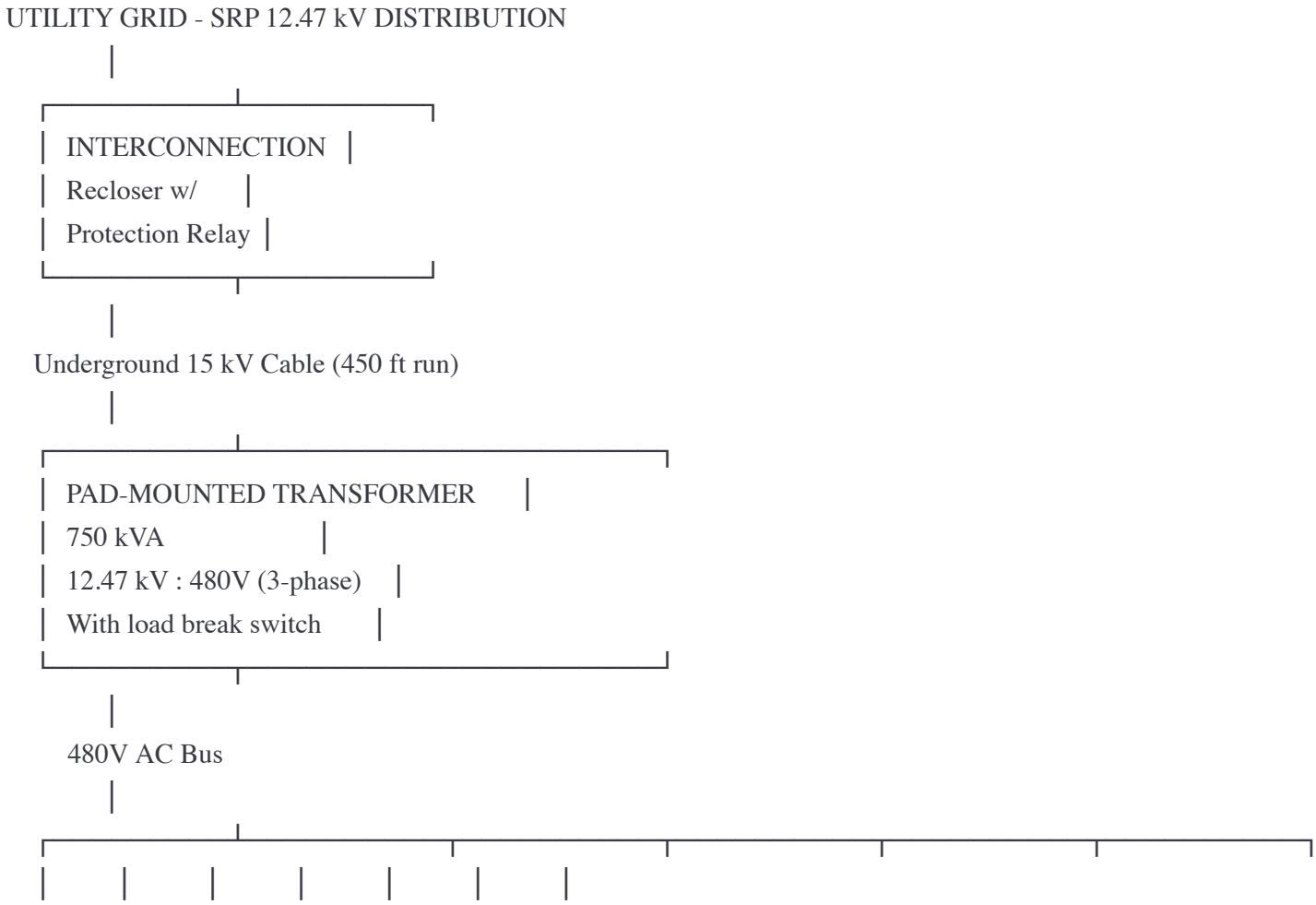
Document 3: Electrical Single-Line Diagram

ELECTRICAL ONE-LINE DIAGRAM

Project: Desert Hills Community Solar Array

Address: 9845 East Brown Road, Mesa, AZ 85207

Prepared by: Desert Power Systems, Inc.



[Total: 48 strings of solar modules]

[Each string: 40-42 modules in series]

[Approximate 330-340 modules per inverter]

[Total: 2,000 modules × 375W = 750 kW DC]

String Configuration:

- Modules per string: 40-42 (depending on string location)
- String voltage: 600-630V DC (at STC)
- String current: 11.2A (module Isc: 14.35A)
- Combiner boxes: 1000V DC rated, 8-position each

AC Collection System:

- Main AC Bus: 480V, 3-phase, 4-wire
- Conductor: 500 kcmil aluminum, THWN-2 in 3" conduit (underground)
- AC disconnects: At each inverter (200A, NEMA 3R)
- Transformer secondary breaker: 1000A

Grounding System:

- Ground grid: 4/0 bare copper, interconnected beneath array
- Ground rods: 10 ft copper-clad, installed at each inverter/combiner location
- Equipment grounding: 4 AWG copper to all equipment
- Lightning protection: Surge arresters at transformer and inverters

Monitoring & SCADA:

- SMA Data Manager with cellular modem
- Production metering: Revenue-grade meter at POI
- Monitoring parameters: AC/DC production, irradiance, temperature, inverter status

Protective Devices:

- Utility-grade protection relay (SEL-351 or equivalent)
- Anti-islanding protection (inverter integral + utility recloser)
- Ground fault protection at combiner boxes
- Arc fault detection in inverters

Document 4: Equipment Specifications Summary

Solar Modules - Jinko Solar Tiger Neo N-type 375W

Model: JKM375N-54HL4-B

Technology: N-type monocrystalline, bifacial, half-cell

Key Specifications:

- Rated Power (Pmax): 375W (front side)
- Bifaciality: 70% (up to 263W additional from rear)
- Module Efficiency: 21.2%
- Dimensions: 71.26" × 44.41" × 1.38"
- Weight: 51.7 lbs
- Cells: 108 cells (2 × 54 half-cut)
- Max System Voltage: 1500V DC
- Operating Temperature: -40°C to +85°C
- Fire Rating: Class C
- Wind Load: Up to 2400 Pa
- Snow Load: 5400 Pa
- Warranty: 30-year performance (84.8% at year 30), 15-year product
- Certifications: UL 1703, IEC 61215, IEC 61730, IEC 61701 (salt mist)

Temperature Coefficients:

- Pmax: -0.30%/°C
- Voc: -0.25%/°C
- Isc: +0.05%/°C

Inverters - SMA Sunny Central 100 CP XT

Model: SC100CP-XT-US

Type: Central inverter, transformer-less

Key Specifications:

- AC Output Power: 100 kW
- Max DC Input: 150 kW
- Max DC Voltage: 1000V
- Operating DC Voltage: 570-850V

- Max DC Current: 240A
- AC Voltage: 480V, 3-phase
- Max AC Current: 120A
- Max Efficiency: 98.5%
- CEC Weighted Efficiency: 98.0%
- Dimensions: 78.7" × 35.4" × 17.7"
- Weight: 1,168 lbs
- Cooling: Forced air (OptiCool)
- Protection Rating: NEMA 3R (outdoor rated)
- Operating Temperature: -13°F to +131°F
- Communication: Ethernet, Modbus, Sunny Portal
- Warranty: 10 years standard (extendable to 20 years)
- Certifications: UL 1741-SA, IEEE 1547-2018, Rule 21, WECC

Safety Features:

- Integrated DC disconnect
- Arc fault circuit interruption (AFCI)
- Ground fault monitoring
- Anti-islanding protection
- Overvoltage/undervoltage protection
- Rapid shutdown compliant

Racking System - Schletter FS Duo Ground Mount

System Type: Fixed-tilt, ballasted foundation

Key Specifications:

- Tilt Angle: 25° (optimized for latitude)
- Module Orientation: Portrait, 2 modules high
- Row Configuration: Single axis row tracking (not used, fixed-tilt selected)
- Foundation: Pre-cast concrete ballasts, 3,000 lb design load per foundation
- Structural Materials: Aluminum alloy 6063-T6 for rails, Hot-dip galvanized steel for posts
- Wind Rating: 140 mph (3-second gust, Exposure C)

- Snow Rating: 30 psf ground snow load
- Height: Module bottom edge 3.5 ft above grade, top edge 10.2 ft
- Certifications: ICC-ES ESR-3794, ASCE 7-16 compliant

Foundation Details:

- Ballast blocks: No ground penetration required
- Soil bearing pressure: <250 psf (suitable for most soil types)
- Environmental: Minimal site disturbance, reversible installation

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Transformer - Pad-Mounted 750 kVA

Specifications:

- Capacity: 750 kVA
- Primary: 12.47 kV, 3-phase
- Secondary: 480V, 3-phase, 4-wire
- Impedance: 5.5%
- Cooling: OAAN (Oil-immersed, air-cooled)
- Tap Changer: ±2.5%, ±5% (no-load)
- BIL Rating: 95 kV
- Sound Level: 56 dBA at rated load
- Enclosure: NEMA 3R, tamper-resistant
- Oil Containment: 110% capacity secondary containment
- Bushings: Dead-front, 200A loadbreak capable
- Warranty: 5 years
- Standards: IEEE C57.12.00, IEEE C57.12.34

Document 5: Interconnection Application Summary

UTILITY INTERCONNECTION APPLICATION
Salt River Project (SRP)

Application Type: Generation Interconnection - Community Solar

Application Date: September 22, 2025

Application Date: September 22, 2025
SRP Application #: GEN-2025-0847

Generator Information:

- Project Name: Desert Hills Community Solar Array
- Generator Type: Solar Photovoltaic
- Generator Capacity: 750 kW DC / 600 kW AC
- Expected Annual Output: 1,425,000 kWh
- Interconnection Voltage: 12.47 kV
- Interconnection Location: Existing distribution pole, East Brown Road

Point of Interconnection:

- Existing SRP Circuit: F24-108
- Feeder Capacity: 12.47 kV, capacity available for 600 kW generation
- Distance from POI to project: 450 feet
- Interconnection Equipment: New recloser with protection relay, underground cable

Technical Studies Required:

- Initial screening: Passed (< 15% of feeder capacity)
- Supplemental review: In progress
- System impact study: Not required (per initial screening)

Protection & Control:

- Anti-islanding: Inverter-based + utility recloser
- Voltage regulation: Inverter reactive power control (0.95 leading to 0.95 lagging)
- Frequency response: Per IEEE 1547-2018
- Communications: DNP3 protocol to SRP SCADA

Insurance Requirements:

- General Liability: $2,000,000 per occurrence / $4,000,000 aggregate
- Additional Insured: SRP named as additional insured
- Proof of Insurance: Provided with application

Expected Timeline:

- Application review: 30-45 days
- Construction authorization: November 2025
- Interconnection agreement execution: December 2025
- Parallel operation: June 2026

Document 6: Land Lease Agreement Summary

GROUND LEASE AGREEMENT (SUMMARY)

Lessor: City of Mesa
Lessee: Southwest Renewable Energy Partners, LLC
Property: Desert Hills Community Park, 5.1 acres
Term: 25 years with option to extend 10 years
Commencement: Upon permit issuance

Lease Payments:

- Year 1-5: $8,000/year
- Year 6-10: $9,000/year
- Year 11-15: $10,000/year
- Year 16-20: $11,000/year
- Year 21-25: $12,000/year
- Extension period: Fair market rate

Additional Considerations:

- Property tax responsibility: Lessee (improvement value only)
- Maintenance: Lessee responsible for all array maintenance
- Decommissioning bond: $75,000 (escrow or letter of credit)
- Vegetation management: Native desert landscaping maintained by lessee
- Public access: Site not open to public; educational signage provided at park entrance

Property Owner Authorization:
The City of Mesa authorizes Southwest Renewable Energy Partners, LLC to:

- Apply for all necessary permits
- Construct, operate, and maintain the solar array

- Interconnect with utility infrastructure
- Access the property for O&M activities

Authorized signatures on file with City Clerk.

Document 7: Environmental & Permitting Checklist

ENVIRONMENTAL COMPLIANCE CHECKLIST

✓ **Biological Survey:** No threatened or endangered species present
✓ **Cultural Resources:** Archaeological survey completed, no sites identified
✓ **Wetlands/Waters:** No jurisdictional waters present
✓ **Stormwater:** AZPDES General Permit for Construction (Notice of Intent filed)
✓ **Air Quality:** No dust control permit required (minimal grading)
✓ **Noise:** Construction noise exempt (daytime construction only)
✓ **Glare Analysis:** FAA Form 7460-1 not required (< 200 ft AGL, no airport impacts)
✓ **Fire Department:** Coordination letter obtained, knox box for gate access
✓ **Zoning:** Conditional Use Permit approved by Mesa Planning & Zoning (August 2025)

Required Submittals with Permit Application:

1. ✓ Site Plan (civil engineered)
2. ✓ Electrical One-Line Diagram
3. ✓ Equipment Specification Sheets (modules, inverters, transformer)
4. ✓ Structural Calculations (racking system, wind/snow loads)
5. ✓ Geotechnical Report
6. ✓ Interconnection Agreement (SRP)
7. ✓ Land Lease Agreement
8. ✓ Stormwater Pollution Prevention Plan (SWPPP)
9. ✓ Decommissioning Plan
10. ✓ Proof of Insurance

Complete equipment specification sheets, installation manuals, structural calculations, and all supporting documentation available in full submittal package.